

the start time of the energy transfer that the ESA has previously suggested it would use.

Washing machine example: A Washing Machine may have been set to start a wash cycle at 9pm when the variable tariff normally reduces.

However, the EMS is aware that a grid event has occurred, making it cheaper to run the cycle at a later time, but the washing machine is not aware of this.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the washing machine has a power profile suggesting it will start the wash cycle at 9pm, but the EMS now knows that the grid event means it will be cheaper to delay the start until 11pm.

The EMS can then optimize the cost by asking the washing machine to delay starting the wash cycle until 11pm.

It does this by sending a [StartTimeAdjustRequest](#) to the washing machine to request delaying the start of the washing cycle.

9.2.4.5. Pausable Feature

ESAs which support the Pausable feature, allow an EMS to recommend a pause in the middle of a forecast power profile that the ESA is currently using.

Washing machine example: A Washing Machine is in operation, and starting its water heating step.

However, the EMS becomes aware from the smart meter that the total home load on the grid is close to exceeding its allowed total grid load.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the washing machine has a power profile suggesting its current step in the wash cycle is using power to heat the water, but that this step can be paused.

The EMS can then reduce the grid load by asking the washing machine to pause the wash cycle for a short duration.

It does this by sending a [PauseRequest](#) to the washing machine to request pausing the current step of the forecast power usage for a period to allow other home loads to finish before resuming the washing cycle.

9.2.4.6. ForecastAdjustment Feature

ESAs which support the Forecast adjustment feature, allow an EMS to recommend a change to the start, duration and/or power level limits of the steps of the power profile that the ESA has previously suggested it would use.

Heat pump and Solar PV example: A heat pump may have the ability to heat hot water as well as heating the home. The heat pump scheduling system may have determined that the home will be unoccupied during the day, or that the indoor temperature is above the set-point and so it knows that it will not need to heat the home.